PROBLEM STATEMENT 26013 · MINISTRY OF RURAL DEVELOPMENT, DOLR · SMART AUTOMATION

Every word in PS 26013 that carries weight

The statement, reproduced in full and marked up. Six colours separate the language that **binds** you from the language that **protects** you — because the permissive words are worth as much as the mandatory ones, and they are the half most teams read straight past. **Tap a colour to isolate it.**

PROBLEM STATEMENT ID	26013
PROBLEM STATEMENT TITLE	Automated lntegration and lntelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management. **Reproduce the title exactly as filed** The published statement writes a lowercase **l** where a capital **I** belongs — "Integration", "Intelligent", "Imagery", "AI" for Al — and mid-word capitals in "incorPorate" and "footPrint". **That is SIH's own typography, carried from the source document.** Quote it verbatim on the title slide; a tidied title stops matching the PS you filed against. Everywhere else, write it correctly.
DESCRIPTION	BACKGROUND Urban land administration and cadastral management involve integration of multiple spatial and non-spatial datasets generated from various departments, agencies, and survey mechanisms. Under modern land governance programmes such as the NAKSHA Programme, large volumes of geospatial data are being generated through drone surveys, Orthorectified lmagery (ORl), DSM/DTM datasets, Ground Truthing (GT), GNSS surveys, municipal records, utility databases, and revenue land records.

At present, harmonization and integration of these datasets largely depend on manual GIS workflows, which are time-consuming and prone to errors. With increasing availability of AI, GeoAI, and automated spatial processing technologies, there is significant scope for development of an intelligent system capable of automatically integrating and synchronizing multi-source geospatial datasets with feature-extracted cadastral data.

> **The pitch is written for you here**
>
> Three words carry the whole business case: **manual**, **time-consuming**, **prone to errors**. That is DoLR's stated pain, so those are the three your impact slide must answer. Note what the statement never asks for: it does not ask you to be *fast*, and it does not ask you to be *right* in some absolute sense — it asks you to remove the manual step without introducing error. A platform that automates the reconciliation and then **shows why each decision was made** answers both halves at once.

DESCRIPTION

The proposed solution should develop an AI-enabled geospatial integration platform capable of automatically integrating, harmonizing, validating, and synchronizing multiple land-related datasets with AI-generated feature extraction outputs.

> **The single sentence a jury can disqualify you on**
>
> Four verbs, and they are not synonyms. **Integrating** is moving data together. **Harmonizing** is making it agree. **Validating** is proving it agrees. **Synchronizing** is keeping it agreeing over time. Most submissions build the first and gesture at the rest. Be able to point at a module for each one, and say plainly which of the four is a design rather than a deployment. The clause also says **"with AI-generated feature extraction outputs"** — a GeoAI layer is not optional decoration here, it is a named input.

THE SYSTEM SHOULD SUPPORT INTEGRATION OF

- Drone imagery
- Orthorectified Imagery (ORI)
- DSM/DTM datasets
- Existing cadastral maps
- Revenue records
- Municipal GIS layers
- Utility network data
- Ground Truthing (GT) datasets
- GNSS/CORS survey data
- Building footPrint datasets

> **Count them — there are ten, and each one is a checkbox**
>
> Ten named inputs, and **an input with no real dataset behind it is an input with no evidence**. A jury reads this list as a checklist. For each, be ready to name the actual file, its authority and its licence — not a connector that "would accept" it. Where a real Indian dataset does not exist openly, say so and show the closest real one you did use. A stated gap survives a question; a silent one does not.

THE SOLUTION SHOULD INCORPORATE

- AI/ML-based spatial matching algorithms
- Automated topology correction
- Intelligent attribute mapping
- Geo-referencing and coordinate transformation engine
- Change detection mechanisms
- Spatial conflict resolution framework
- Confidence scoring for integrated outputs

> **Seven capabilities, and the last two are the whole problem**
>
> The first five are engineering. **Spatial conflict resolution** and **confidence scoring** are the two that decide whether this is a land-records platform or an ETL pipeline with good marketing. The statement puts them last but they are what "harmonize" actually means: when four sources disagree about one boundary, something must decide, and something must say how sure it is. Build those two first and the other five become their inputs.

EXPECTED SOLUTION

The expected outcome is development of an intelligent geospatial integration framework capable of automatically harmonizing multi-source land-related datasets with AI-generated feature extraction outputs. The final solution should:

- Reduce manual GIS integration efforts
- Improve accuracy and consistency of urban land records
- Enable seamless inter-departmental spatial data exchange
- Accelerate cadastral finalization processes
- Improve interoperability of urban land information systems
- Support standardized digital land governance

> **These six are the marking scheme**
>
> Outcomes, not features — and five of the six are institutional rather than technical. **"Seamless inter-departmental exchange"** and **"interoperability"** are answered by a standard, not by an API you invented: if a revenue officer can open your endpoint in QGIS with no plugin, that clause is met and demonstrable in ten seconds. **"Accelerate cadastral finalization"** is the one to quantify — finalisation is blocked by disputed parcels, so the honest measure is how many disagreements you close automatically and how well you rank the ones you cannot.

SUGGESTED TECHNOLOGIES

Artificial Intelligence (AI) · Machine Learning (ML) · GeoAI · GIS & Web-GIS · Spatial Databases · ETL Automation · Computer Vision · Cloud Computing · Spatial Analytics · API Integration Frameworks

> **The word "suggested" is doing a lot of work**
>
> This list is **latitude, not obligation**. Nothing here says cloud, and nothing here says deep learning. If your defensible answer is boosted trees on a laptop and an on-premises deployment because a district office has neither GPUs nor bandwidth, that is *within* the statement and it is a stronger answer than a cloud diagram. Use the list to show breadth where you genuinely have it, and say why you declined the rest.

ORGANIZATION	Ministry of Rural Development

DEPARTMENT	Dept of land resources (DoLR)
CATEGORY	Software
THEME	Smart Automation

What binds you

Four verbs — **integrate, harmonize, validate, synchronize** — over **ten named datasets**, incorporating **seven named capabilities**, judged against **six stated outcomes**. That is 27 checkable clauses, and the AI-generated feature-extraction input is named twice, so it is not decoration.

The only sentence written as a hard constraint is the platform definition itself. Everything else is *should*, and *should* is answerable with a stated limitation.

What protects you

"Suggested Technologies" — the ten-item stack is explicitly suggested. You owe the outcomes, not the toolchain.

"such as" before NAKSHA and before the dataset list — the programme is an example of the context, not a required integration.

"significant scope for development of" — the background frames this as an opportunity, which is what lets you scope one AOI honestly instead of promising a nation and demonstrating nothing.

Reproduced from the published problem statement · marked up for GEOVAX, Team Perpetually Curious · no wording altered